Raspberry Pi 4 B — running one test on the board, step by step

This page runs one µOS++ test on a Raspberry Pi 4 B (BCM2711, 4× Cortex-A72) through OpenOCD, by hand, for the 64-bit (AArch64) port. It does exactly what the test scripts do (`micro-os-plus-iii-aarch64/test/boards/rpi4b/hw.sh`, which calls `micro-os-plus-iii/test_smpl/run-hw.sh`), written out as plain commands: reset, load, run, watch, end.

The probe is a Raspberry Pi **debugprobe** (a Pico) used as a **CMSIS-DAP v2** adapter in **JTAG** mode, through OpenOCD (xPack 0.12.0-7.1). The Pico runs an updated firmware with optimized CMSIS-DAP/JTAG code (it still reports `FW Version = 2.0.0`), which loads about 38× faster than the stock debugprobe firmware (see “Load speed” below).

What happens, in short

1. **Reset.** OpenOCD connects, halts core 0, and writes the BCM2711 PM watchdog (`0xfe100024` = `0x5a000001`, then `0xfe10001c` = `0x5a000020`). The whole SoC resets and boots again from the SD card. OpenOCD loses the link and exits.
2. **Wait 10 s** for the firmware to boot and switch the JTAG pins on again. Core 0 runs the firmware stub; cores 1–3 wait in the stub’s spin-table loop.
3. **Connect and halt core 0** (it stops at EL2, MMU off). Cores 1–3 are not halted (see step 6).
4. **Semihosting on**, on all four cores.
5. **Load.** `load_image` copies the test’s ELF into RAM at `0x80000`.
6. **Run.** Core 0’s PC is set to `0x80000` (`_start`) and core 0 is resumed. Cores 1–3 go on waiting in the firmware’s loop until the kernel writes their spin-table slot.
7. **Watch.** The test’s console is the UART. The verdict line (`RESULT: PASS|FAIL|SKIP`) also goes out through semihosting, where the runner sees it.
8. **End.** As soon as the verdict appears, the runner stops OpenOCD. The next test starts again at step 1; no power cycle is needed.

The hardware

What	Where
JTAG	the Pi’s GPIO 22–27 (ALT4: TRST, RTCK, TDO, TCK, TDI, TMS), enabled by `enable_jtag_gpio=1` in `config.txt`
Probe	Pico debugprobe, USB `2e8a:000c`, CMSIS-DAP v2, firmware 2.0.0
Console	the Pi’s PL011 UART (GPIO 14/15), 115200 8N1, here on a CH340 adapter at `/dev/ttyUSB0`
SD card	the boot card: firmware, `config.txt` and a `kernel8.img`. The test’s `config.txt` is `micro-os-plus-iii-aarch64/test/boards/rpi4b/config.txt`

The hwd tests that use the SD card (`sd_test`, `smp-mat-sdcard-test`, `smp-num-test`, `smp-pipeline-test`) mount the card’s **existing FAT32 boot partition** and write only under a `tests/` folder. They never format the card and never touch the boot files.

Step 0 — set the paths

```
W=/home/dan/Desktop/micro-os-plu-iii-smp-pr-github
OCD_DIR=$HOME/.local/xPacks/@xpack-dev-tools/openocd/0.12.0-7.1/.content
OCD=$OCD_DIR/bin/openocd
SCRIPTS=$OCD_DIR/openocd/scripts
BOARD=$W/micro-os-plu-iii-aarch64/test/boards/rpi4b
CFG=$BOARD/openocd-cmsis-dap-rpi4b.cfg
C=aarch64-rpi4b-cmake-gcc-debug
TESTS=$W/micro-os-plus-iii/tests
ELF_DIR=$TESTS/build/$C/platform-bin/port-tests/test
```

Step 1 — build the tests

```
cd $TESTS
xpm run install --config $C     # first time only
xpm run prepare --config $C
xpm run build   --config $C
```

Each test gives `<app>-hwd` (for the board) and `<app>-qemu`. The `-hwd` name is a symlink to `<app>-hwd.elf`.

Optional: semihosting prints (`SEMI_PRINTS`)

By default (`SEMI_PRINTS=0`) a board image sends only its verdict lines through semihosting; the whole console is on the UART. To load the probe with semihosting traffic, build the hwd images with `SEMI_PRINTS=1`: every console line printed on **core 0** then also goes through semihosting (and appears in the OpenOCD output). Lines from cores 1–3 stay on the UART only, because OpenOCD 0.12 does not resume a core other than 0 after its semihosting call on this board (see “If something goes wrong”).

```
cmake -DSEMI_PRINTS=1 $TESTS/build/$C     # back to the default: -DSEMI_PRINTS=0
xpm run build --config $C
```

It is slow: OpenOCD reads each line from the board one byte at a time, about 0.9 s per line. smp_mem_test (~300 lines from core 0) passes in 337 s instead of 54 s, smp_mem_inter_test (~480 lines) in 621 s instead of 186 s. The qemu images always mirror everything.

Step 2 — choose the test

```
ls $ELF_DIR/*-hwd
APP=smp-mat-test
ELF=$ELF_DIR/$APP-hwd
```

What matters for the load time is the size of the loadable segment (`FileSiz` of the `LOAD` line), not the size of the ELF file (that one also holds the debug information, which is never sent to the board):

```
$TESTS/build/$C/xpacks/.bin/aarch64-none-elf-readelf -lW $ELF | grep LOAD
```

With the optimized probe firmware the load runs at about **39.5 KB/s** (smp_test4: 70,296 bytes in 1.78 s, measured). The stock debugprobe firmware did about 1 KB/s (see “Load speed” below). With the PIO probe firmware (JTAGprobe_smp_pio) and `UOS_HW_ADAPTER_KHZ=15000` the load runs at

about **58–59 KB/s**, about 1.5× faster than this column. The new column is that rate applied to each test's size; only smp_test4 was timed:

Test	Loaded bytes	Load (optimized firmware)	Load (stock firmware)
cmsis-os-validator	151,640	~3.8 s	~2.4 min
mutex-stress	94,488	~2.4 s	~1.5 min
rtos-apis	228,136	~5.8 s	~3.5 min
sd_test	83,512	~2.1 s	~1.3 min
smp-mat-sdcard-test	93,736	~2.4 s	~1.5 min
smp-mat-test	77,608	~2.0 s	~1.2 min
smp-num-test	91,160	~2.3 s	~1.4 min
smp-pipeline-test	146,472	~3.7 s	~2.3 min
smp-pro-cons-test	88,968	~2.3 s	~1.4 min
smp_test0 … smp_test4	65,656 … 74,744	~1.7 … 1.9 s (smp_test4: 1.78 s measured)	~1.1 min

Step 3 — open the serial console

In a second terminal, and leave it open:

```
tio -b 115200 /dev/ttyUSB0
```

The test's whole output appears here once it is started (step 6). On the board, only the verdict lines go through semihosting (with `SEMI_PRINTS=1`, core 0's lines too: step 1).

Step 4 — reset the board

```
pgrep -x openocd && echo "an OpenOCD still holds the probe: pkill -x openocd"

$OCD -s $BOARD -s $SCRIPTS -f $CFG -c init \
  -c "targets bcm2711.cpu0; halt; \
      catch {mww 0xfe100024 0x5a000001}; \
      catch {mww 0xfe10001c 0x5a000020}; \
      shutdown"
```

`0x5a` is the PM password; `0xfe100024` (PM_WDOG) gets a timeout of 1 tick, and `0xfe10001c` (PM_RSTC) gets "full reset".

The address is the **virtual** one, not `mww phys`. With the MMU and D-cache on (a test was running), `mww phys` makes OpenOCD clean the caches by set/way over JTAG first, which never finishes at this link speed.

What a good reset prints: the TAP is found, and then the link drops when the SoC resets:

```
Info : JTAG tap: bcm2711.cpu tap/device found: 0x4ba00477 (mfg: 0x23b (ARM Ltd), part: 0xba00,
ver: 0x4)
...
Error: Invalid ACK (0) in DAP response
Error: [bcm2711.cpu0] Polling failed, trying to reexamine
```

Those last errors are the proof that the reset happened. If the output has no `tap/device found` line, or a Tcl `Traceback`, the watchdog was **not** written: fix that first. The runner checks this and stops with `watchdog reset FAILED`.

The same reset is in `$BOARD/reset.cfg`:

```
$OCD -s $BOARD -s $SCRIPTS -f $BOARD/reset.cfg
```

Step 5 — wait for the board to boot

```
sleep 10
```

Step 6 — load the test and start it

Write the session's commands to a file (this is what the runner generates, in `$ELF_DIR/.hw-logs/<app>.cfg`):

```
cat > /tmp/run-$APP.cfg <<CFG
echo "--- stage: halt cores 0 ---"
targets bcm2711.cpu0; halt

echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
# Per-target: a core whose semihosting is off stalls at a debug halt the first
# time it traps, so every core needs it, not just core 0.
foreach core {0 1 2 3} { targets [format {bcm2711.cpu%d} \$core]; arm semihosting enable }
if {[info procs rpi4b_enable_hde] ne ""} { rpi4b_enable_hde }

echo "--- stage: load_image $ELF ---"
targets bcm2711.cpu0
load_image $ELF

echo "--- stage: resume cores 0 at 0x80000 ---"
targets bcm2711.cpu0
reg pc 0x80000
echo "  start PC core 0 = [reg pc]"
resume
CFG
```

Only **core 0** is halted, loaded through and resumed. Cores 1–3 are never halted: they keep running in the firmware's spin loop until the kernel writes their spin-table slot. Halting them too looks harmless but is not: a resume of core 0 restarts the other cores through the CTI (cross-trigger), so OpenOCD never resumes them itself, and a core it halted at the start later halts with the registers OpenOCD cached at that first halt. OpenOCD then cannot serve that core's semihosting call, the core stays halted, and the test stalls.

`0x80000` is `_start`, the ELF's entry point (`aarch64-none-elf-readelf -h $ELF | grep Entry`).

Then run OpenOCD with it, and **leave it running**: it serves the test's semihosting calls.

```
$OCD -s $BOARD -s $SCRIPTS -f $CFG -c init -f /tmp/run-$APP.cfg 2>&1 | tee /tmp/run-$APP.log
```

What it prints before the load, all normal:

```
Info : JTAG tap: bcm2711.cpu tap/device found: 0x4ba00477 ...
Error: JTAG-DP STICKY ERROR
```

```
Error: Failed to read memory at 0x80410088        (and 0x80510088, 0x80610088, 0x80710088)
bcm2711.cpu0 halted in AArch64 state due to debug-request, current mode: EL2H
cpsr: 0x000003c9 pc: 0x200
```

The `Failed to read memory at 0x80n10088` lines

One pair of `JTAG-DP STICKY ERROR` / `Failed to read memory` lines per core, always the same four addresses. They are harmless: the test is not affected, and the runner ignores exactly these (`UOS_HW_IGNORE_ERRORS` in `hw.sh`).

Address	Core	Register
`0x80410088`	0	`EDSCR` (debug base `0x80410000` + `0x088`)
`0x80510088`	1	`EDSCR` (`0x80510000` + `0x088`)
`0x80610088`	2	`EDSCR` (`0x80610000` + `0x088`)
`0x80710088`	3	`EDSCR` (`0x80710000` + `0x088`)

`EDSCR` is the core's External Debug Status and Control Register. What happens, for each core, in this order (OpenOCD `src/target/target.c`, `target_examine_one`):

1. OpenOCD fires the core's `examine-start` event. The config hooks `rpi4b_unwedge <debug base>` there, whose first action is to read `EDSCR` through the APB-AP.
2. Only then does OpenOCD examine the core (`aarch64_examine_first`), and the first thing that does is write `OSLAR = 0`: it unlocks the core's debug **OS Lock**.

The board has just been reset by the watchdog (Step 4), and a reset sets the OS Lock of every core (OpenOCD reports it as `external reset detected`). While the OS Lock is set, the core refuses an external read of `EDSCR`; the JTAG-DP flags the refused access as a sticky error, and OpenOCD prints the two lines.

Why it does no harm:

- `rpi4b_unwedge` reads `EDSCR` inside `catch`: the failed read makes it return, and the session goes on. After a fresh reset there is nothing to unwedge anyway.
- The same `EDSCR` read at `examine-end` (`rpi4b_enable_core_hde`), after OpenOCD has unlocked the core, gives no error – the proof that the OS Lock is the cause.
- `rpi4b_unwedge` matters for the reset session of Step 4, which connects to a board that has *not* been reset yet: there the OS Lock is clear and a core left stuck by a killed session can be recovered.

To silence them, `rpi4b_unwedge` would have to write `OSLAR = 0` (`<debug base> + 0x300`) itself before its first read – the same write OpenOCD does. The config does not do that today; the lines are only cosmetic.

What the config file `$CFG` sets up, besides the board:

Line	Why
`source [find interface/cmsis-dap.cfg], transport select jtag`	the probe, in JTAG mode
`source [find board/rpi4b.cfg]`	OpenOCD's BCM2711 target: `bcm2711.cpu0..3`, `bcm2711.dap`, `bcm2711.ap` (not grouped: `USE_SMP` is 0), linked by their CTIs
`adapter speed 4000`	4 MHz TCK
`gdb port disabled`	no GDB, OpenOCD only

Line	Why
`rpi4b_unwedge` on every `examine-start`	clears a core's stuck debug state (EDSCR ERR/MA, TX/RX full) left by an earlier session
`bcm2711.dap memaccess 0` on every `examine-end`	no idle TCKs after each access: the load is 1.74× faster (see below). It has to be set after `init`, and every core's examine resets it to 10, so it is set again after each one

Step 7 — watch the result

On the UART (`tio`), the test's own output, ending with the verdict:

```
RESULT: PASS
```

In the OpenOCD output, only the verdict line arrives (through semihosting), plus core 0's console lines in a `SEMI_PRINTS=1` build. The runner waits for one of:

In the OpenOCD output	Meaning
`RESULT: PASS`	passed
`RESULT: SKIP`	skipped
`RESULT: FAIL`, `[FAIL]`, `FAIL:`, `FATAL:`	failed
OpenOCD exits	OpenOCD died
`Invalid ACK`, `Polling failed`, `Examination failed`	debug link lost
any other `Error:` line	OpenOCD error
nothing within the time limit (1800 s)	timeout

Step 8 — end OpenOCD

When the verdict is there, stop OpenOCD (Ctrl-C, or):

```
pkill -x openocd
```

Stopping it while it is talking to the probe prints `Error: error handling USB events: System call interrupted (perhaps due to signal)`: that is the stop itself, not a problem.

The board is now running (or stopped in) the finished test. The next test starts again at step 4: the watchdog reset puts the board back in a clean state, so there is **no power cycle** between tests.

The same, with the scripts

One test, with its result in CTest (reset, wait, load, run, verdict, end):

```
cd $TESTS
export PATH=$PWD/build/$C/xpacks/.bin:$PATH
ctest --test-dir build/$C -R '^aarch64-rpi4b-smp-mat-test-hwd$' -V
```

Or through xpm, which builds first:

```
xpm run test-smp-mat-test-hwd --config $C
```

All hwd tests, one after another (each one resets the board first):

```
xpm run test-hwd --config $C          # prepare + build, then ctest -V -L hwd
```

The port's script directly, without CTest:

```
$BOARD/hw.sh list                      # what is built, with time limits
BUILD=$TESTS/build/$C/platform-bin/port-tests $BOARD/hw.sh smp-mat-test 1800
```

Files the runner leaves in `$ELF_DIR/.hw-logs/`:

File	What
`<app>.reset.log`	OpenOCD output of the reset session (step 4)
`<app>.cfg`	the commands of step 6
`<app>.log`	OpenOCD output of the test session (steps 6–7)

Settings (environment variables) the scripts take:

Variable	Default	What
`UOS_HW_RESET_WAIT`	10	seconds between the watchdog reset and the connect
`UOS_HW_CORES`	`0`	cores that are halted, loaded through and resumed
`UOS_HW_SEMIHOST_CORES`	`0 1 2 3`	cores with semihosting enabled (without halting them)
`UOS_HW_ADAPTER_KHZ`	(config: 4000)	JTAG clock, set after `init`; 15000 tested with the PIO probe firmware (see "Load speed")
`UOS_HW_PRELOAD`	the two watchdog writes	the reset commands of step 4
`UOS_HW_IGNORE_ERRORS`	`JTAG-DP STICKY ERROR|Failed to read memory at 0x80[4-7][0-9a-f]+$`	`Error:` lines that do not end the run
`OPENOCD`	newest xPack OpenOCD	the OpenOCD binary

From VS Code

The VS Code project (`RPI4B/64/rpi4b`) uses the same steps with its own config, `rpi4b-cmsis-dap.cfg` (same `rpi4b_unwedge` and `memaccess 0` events):

```
./reset-board.sh 30      # step 4 + a 30 s wait (step 5)
```

and then the "Debug Pi 4B" launch does steps 6–7 through GDB (`monitor halt`, `load`, `monitor arm semihosting enable`).

Load speed

With the **optimized probe firmware** (the Pico reflashed with new optimized CMSIS-DAP/JTAG code), measured on this setup at `adapter speed 4000`, with `bcm2711.dap memaccess 0`: smp_test4 (70,296 loaded bytes) in **1780 ms**, about **39.5 KB/s** (38.6 KiB/s), timed around `load_image` alone in

OpenOCD (`clock milliseconds` before and after). That is about **38×** the stock firmware's ~1.05 KB/s, and the whole test ran and passed (29.6 s in CTest, the 15 s reset wait included).

Probe firmwares compared

Measured on this setup, smp_mem_test and smp_mem_inter_test hwd with `SEMI_PRINTS=1`, all **PASS** (`load_image` timed alone; whole test in CTest, 10 s reset wait included). The firmwares are at github.com/dan-maio/JTAGprobe_smp and github.com/dan-maio/JTAGprobe_smp_pio.

Probe firmware	`adapter speed`	smp_mem_test load	smp_mem_test total	smp_mem_inter_test load	smp_mem_inter_test total
JTAGprobe_smp, 64-byte DAP packets	4000	1,994 ms	337.4 s	2,143 ms	620.5 s
JTAGprobe_smp, 512-byte DAP packets	4000	1,528 ms	308.1 s	1,644 ms	561.8 s
JTAGprobe_smp_pio (PIO JTAG, TMS in the PIO stream)	10000	1,286 ms	291.5 s	—	—
JTAGprobe_smp_pio	15000	1,273–1,287 ms	282.9–285.1 s	1,399 ms	537.4 s

- With the CPU bit-banging firmwares the TCK clock hardly matters (the USB round trip per packet is the limit). With the PIO firmware it does: the JTAG work per 512-byte packet goes from 581 µs (CPU, 4 MHz) to 292 µs (PIO, 10 MHz); 15 MHz gives the same as 10 MHz, so the probe's CPU, not TCK, is then the limit.
- Most of a whole test with `SEMI_PRINTS=1` is semihosting: OpenOCD reads each string one byte per JTAG access, ~0.9 s per line, whatever the probe.
- The config keeps `adapter speed 4000`; use `UOS_HW_ADAPTER_KHZ=15000` with the PIO firmware.

Why the stock firmware was slow

Measured on this setup with the stock debugprobe firmware 2.0.0, with OpenOCD's packet-level log (`-d4`):

- OpenOCD's CMSIS-DAP driver, in **JTAG** mode, sends raw `DAP_JTAG_Sequence` commands and encodes every TMS bit and every idle clock as its own 2-byte entry. One 32-bit access (a 35-bit DR scan plus idle clocks) fills a whole 64-byte USB packet: 4 bytes of data per packet.
- Each packet waits for its answer: **8.4 ms** per round trip on average (full-speed USB, the probe behind a hub).
- So the TCK frequency hardly matters: about 0.6 KB/s at any clock.

With `bcm2711.dap memaccess 0` (10 idle clocks per access removed), a 4 KB load went from 6.7 s to 3.9 s (1.74×) and read back identical. The rest of the cost was the round trip, and that is what the optimized probe firmware removed (above).

If something goes wrong

Symptom	Cause, fix
`LIBUSB_ERROR_BUSY`, or the probe is not found	another OpenOCD holds it: `pkill -x openocd`
the reset prints `The 'bcm2711.dap memaccess' command must be used after 'init'`	`memaccess` written as a plain line in the config; it must be in the `examine-end` events

Symptom	Cause, fix
the reset session shows no `tap/device found`	no JTAG: the board is off, or the firmware has not enabled the pins (`enable_jtag_gpio=1`), or it is still booting (wait longer: `UOS_HW_RESET_WAIT=30`)
`Error: [bcm2711.cpu1] not halted` at the resume, then OpenOCD exits	a script that halted cores 1–3 and resumes them after core 0: the core 0 resume already restarted them through the CTI. Halt and resume core 0 only (`UOS_HW_CORES=0`)
the output stops, and a core 1–3 shows `halted ... due to breakpoint`, `pc: 0x80`, EL1, registers equal to its first halt	that core was halted at the start of the session: OpenOCD later reads its stale cached registers and cannot serve its semihosting call. Do not halt cores 1–3 (`UOS_HW_CORES=0`)
the UART stops after a line from core 1–3, and the OpenOCD output has that line; OpenOCD keeps polling, that core stays `halted`	the core made a semihosting call: OpenOCD 0.12 prints the string but does not resume a core other than 0 on this board, and every other core then waits on the console lock. Only core 0 may print through semihosting (what `SEMI_PRINTS=1` does)
the run log shows `RESULT:` without `PASS`/`FAIL`	the verdict was printed in pieces; the board's `uart.hpp` must mirror the whole verdict line (`mirror_verdict`) or use a single `puts()`
nothing on the UART after the halt	the load is still running: a few seconds with the optimized probe firmware, minutes with the stock one (see the table in step 2)
`mww phys` never ends	use the virtual address (step 4)